Project Summary: Price Comparison Tool

Project Title

Price Comparison Tool

Objective

The objective of this project is to build a Python-based tool that collects and compares laptop prices, ratings, and product names from multiple e-commerce websites. The system helps users identify price differences for the same or similar laptop models across platforms.

Project Overview

The tool uses web scraping techniques to extract publicly available product information from e-commerce websites such as Amazon and Flipkart. The collected data is cleaned, structured, and stored in a CSV file. A basic data visualization (bar chart) is generated to compare average prices across websites.

Technologies Used

- Python
- Requests
- BeautifulSoup
- Pandas
- Matplotlib

Methodology

1. A search query (e.g., "HP Pavilion Laptop") is sent to each website.
2. HTML content of the search results page is fetched using HTTP requests.
3. BeautifulSoup is used to parse the HTML and locate product listings.
4. Key details such as product name, price, and rating are extracted.
5. Error handling is applied to manage missing or inconsistent data.
6. Data from all websites is combined into a Pandas DataFrame.
7. The final dataset is exported as a CSV file.
8. A bar chart is generated to visualize average price differences.

Output

- CSV file containing product comparison data
- Bar chart showing average laptop prices per website

Ethical Considerations

- Only publicly visible data is scraped
- Less than 50 items are collected per website
- No login or restricted content is accessed

Conclusion

This project demonstrates practical use of web scraping, data processing, and visualization techniques in Python. It provides a simple and effective way to compare product prices across platforms and can be extended to include more websites, products, or advanced analytics.